INCH-POUND

MIL-PRF-49471/3(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X590/U

This specification is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49471(ER).

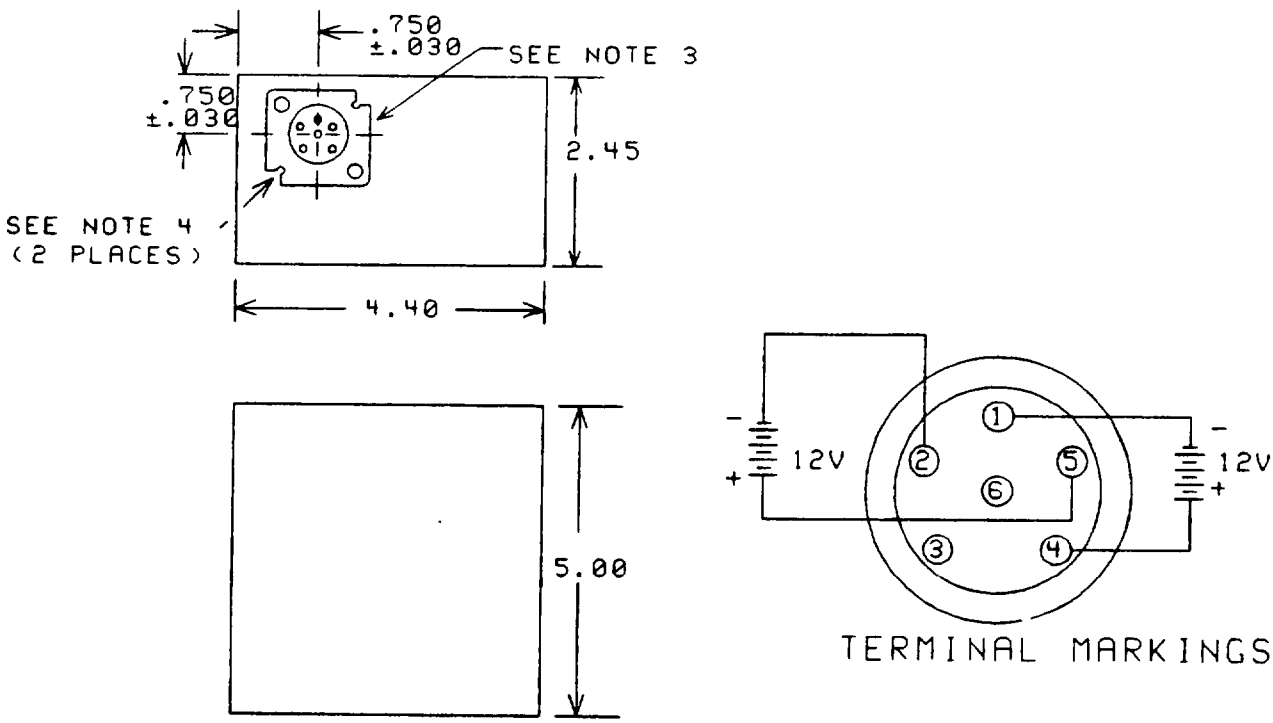


Figure 1. Battery dimensions.

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -0.063, +0.000.

3. The battery terminal shall consist of a miniature circular battery connector per Dwg. SC-C-179492 or equal. Connector shall be mounted in accordance with referenced drawing.

4. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guidepost as shown. Depth of cavity shall be 0.657 inch minimum.

5. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 0.0625 inch below adjacent outside of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 30.50 volts in 24 volt mode
15.25 volts in 12 volt mode

Minimum voltage for capacity: 20.00 volts in 24 volt mode
10.00 volts in 12 volt mode

Charge protection device: Applicable.

Safety feature: Applicable.

Terminals: Six (6) hole socket type with no obstruction of any holes (see Fig. 1).

Weight (maximum): 2.33 pounds (1058 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

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Capacity Test*	Minimum Capacity Requirements in Hours to 20.00 volts in Series, 10.00 volts in Parallel
I & IP	3.4
L & LP	2.2
H & HP	5.6
HT & HTP	5.4
LT & LTP	2.0
IT & ITP	3.0

*NOTE: Sample batteries that are to be assigned to a capacity test shall be equally divided between the designated capacity test and the capacity test ending in P (e.g. I and IP). The odd battery, if present, is assigned to parallel discharge capacity test. For the capacity tests that do not have the P's, the battery's legs shall be discharged in series. For the capacity tests that have the P's, the battery's legs shall be discharged in parallel.

Initial voltage delay: During the I, H, HT, and IT discharge tests of the two (2) 12 volt sections connected in series under the designated loads, initial voltage below the 20.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 20.00 volts cannot exceed a 60 second duration. During the IP, HP, HTP, and ITP discharge tests of the two (2) 12 volt sections connected in parallel under the designated loads, initial voltage below the 10.00 volt minimum voltage cannot exceed a 5.0 second duration. During LP and LTP discharge tests initial voltage below 10.00 volts cannot exceed a 60 second duration.

High temperature device: Applicable.

Complete discharge device: Applicable.

Connector: The connector shall be in accordance with SC-C-179492 or equivalent with the following exception: The durability test described in paragraph 4h shall be performed 50 times with SM-D-687888 or equivalent. Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin.

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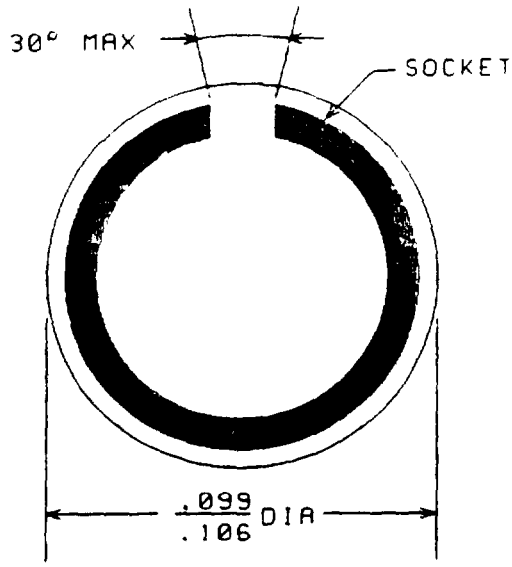


Figure 2. Socket (Top View).

METHOD OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all specified capacity tests are described in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at $70\pm5^{\circ}$F to 2/3 of open-circuit voltage. Each discharged cell shall then be connected in series to nine fresh cells. Each string shall be discharged at 2.0 amperes at $70\pm5^{\circ}$F to 2/3 open-circuit voltage of the nine fresh cells.

Batteries: For all capacity tests that do not end in P, except the H and HT, the batteries shall be discharged in the 24 volt mode by connecting pins 2 & 4 and discharged between pins 1 & 5 at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged in the 24 volt mode as described above, at two amperes for five minutes, followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

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For all capacity tests that end in P, except the HP and HTP, the batteries shall be discharged with the two 12 volt sections connected in parallel at four amperes constant current to zero volts. For the HP and HTP capacity tests the batteries shall be discharged with the two 12 volt sections in parallel at four amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage: Load resistance of 6.0 ohms shall be used between socket holes 1 & 4 and 2 & 5. Voltage shall be 11.70 volts or greater within twenty seconds.

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(Project 6135-A274-03)

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